Catchment-Aware Flood Forecasting System

R25-060

Meet our team…

Perera B.T.S
IT21819360

Amarasekara S.K
IT21340246

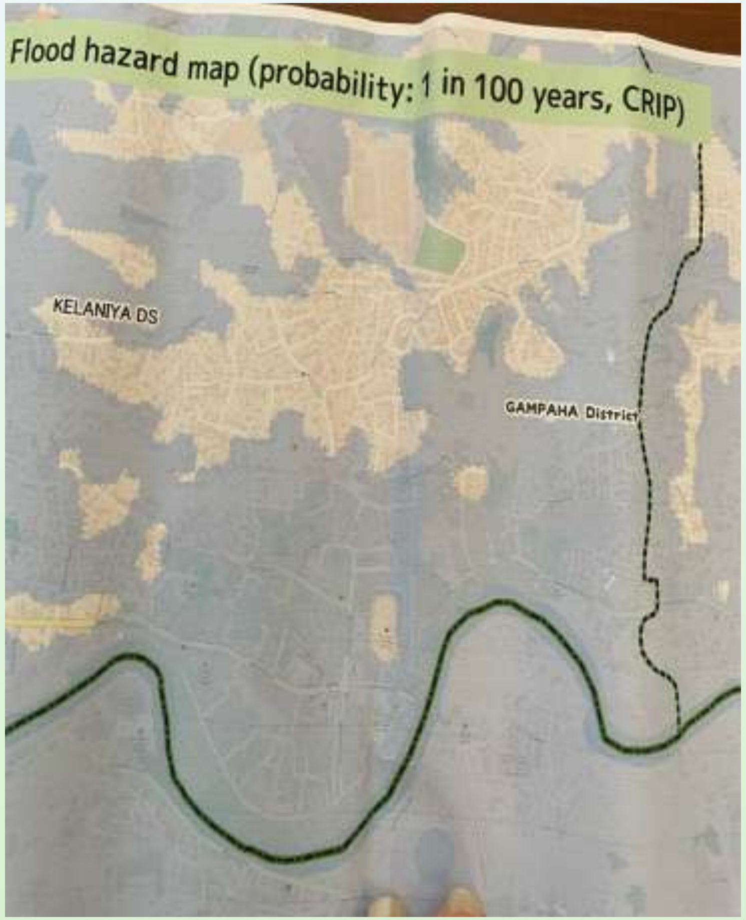
Samaranayake N.A.S.R
IT21816772

Dias A.H.S.G
IT21305214

Introduction

- In Sri Lanka, flooding is a common and destructive phenomenon brought on by excessive rainfall.
- Given the ongoing nature of these weather events, it's crucial for residents to stay informed through official channels and adhere to safety advisories issued by authorities.
- Despite Sri Lanka's advanced technology, AI and IoT are underused for faster, accurate, and actionable flood forecasts.
- The study proposes a Catchment-Aware Flood Forecasting System using XAI, Edge Computing, and IoT to address these issues.

Snaps from the fiel visits



Objectives

Main objective

To develop an **IoT-integrated, edge-computing-based, and Explainable AI-driven flood forecasting system** that eliminates latency issues in real-time water level monitoring and enhances accuracy and speed.

Sub-objectives

1. Flood Impact Prediction and Resource Optimization System with Real-Time Delivery Monitoring
2. Visualization of flood-affected areas and suggesting alternative routes to the users who are affected by flood.
3. An IoT maintenance admin dashboard panel for device management.
4. Provide security guidance for flood-affected individuals to find safe sites and essential supplies.

IT21819360 | Perera B.T.S

- Specialization: Information Technology

Introduction - Background

- **Floods** are a recurring natural disaster, especially in flood-prone regions like Sri Lanka.
- Challenges in current flood management systems
  - Limited predictive capabilities
  - Lack of integration between damage prediction and resource allocation
  - Insufficient real-time data for decision-making.
- Need for a comprehensive flood management solution
  - Integrated flood damage prediction
  - Real-time displacement forecasting
  - Optimized resource allocation strategies

Introduction – Background

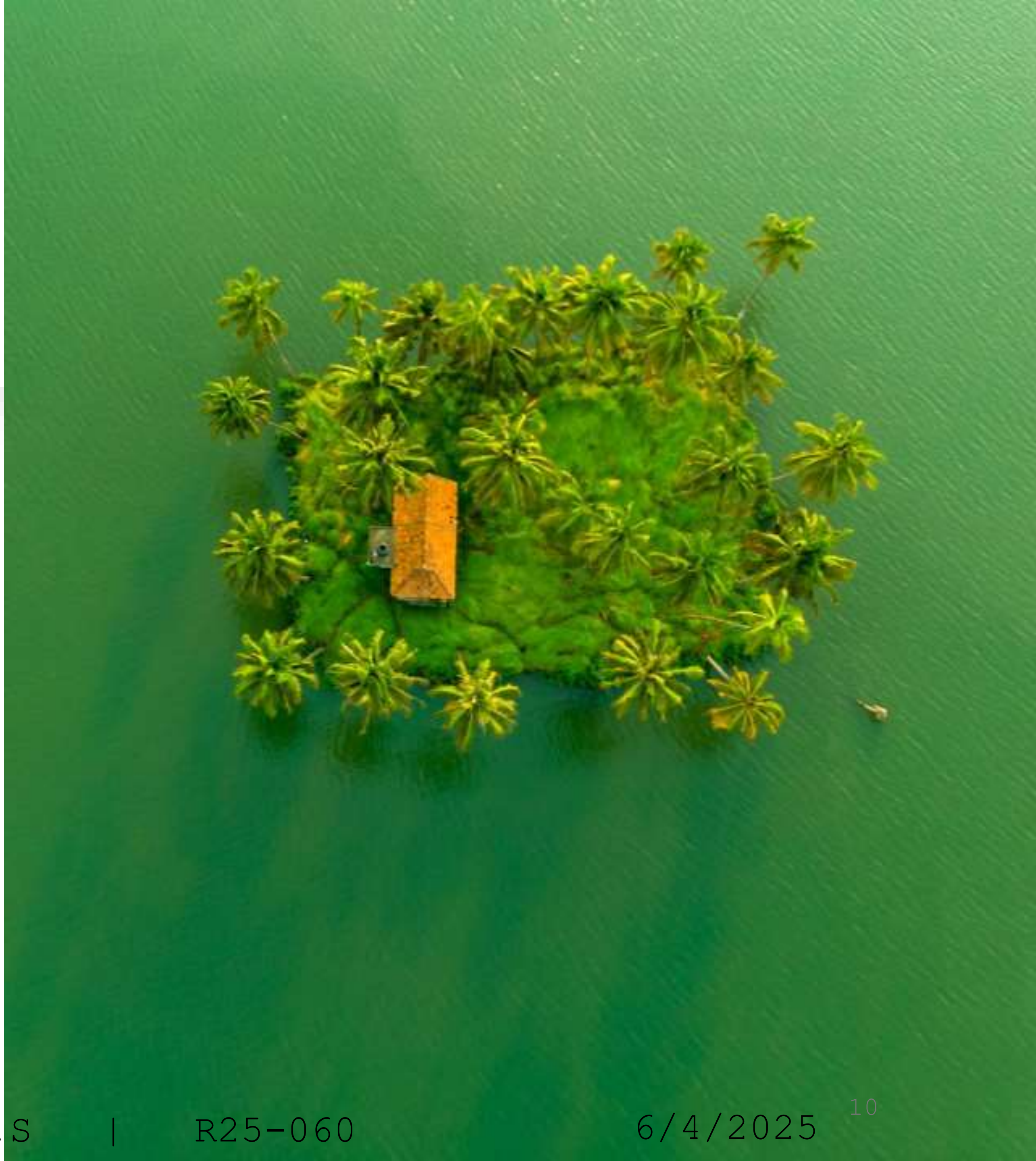
- Main objectives
  - Improve flood damage prediction accuracy
  - Forecast displaced populations based on socio-economic data
  - Optimize resource allocation for efficient disaster response
- System designed for scalability and integration
  - Can be deployed for specific areas (e.g., Kelaniya River Basin)
  - Scalable to cover other flood-prone areas in Sri Lanka
- Real-time actionable insights for stakeholders
  - Donors, disaster management agencies, area managers, and safe site managers
  - Enhanced decision-making and flood response planning

Research Gap

Aspect	Existing system	Gap Identified
Socio-Economic Factors	Not fully integrated into damage and displacement predictions.	Our system incorporates socio-economic data for more accurate predictions.
Real-Time Data Integration	Uses static data (historical records) for predictions.	Lack of real-time flood data integration for dynamic predictions.
Displacement Forecasting	Predicted based on flood extent, often static.	Our system dynamically adjusts displacement forecasts based on real-time data and socio-economic factors.
Resource Allocation Optimization	Static resource allocation methods.	Uses Linear Programming (LP) for real-time optimization based on actual resource needs.
Stakeholder Communication	No real-time updates or interactive communication tools for stakeholders	Real-time updates for donors, managers, and responders on resource

Research Question

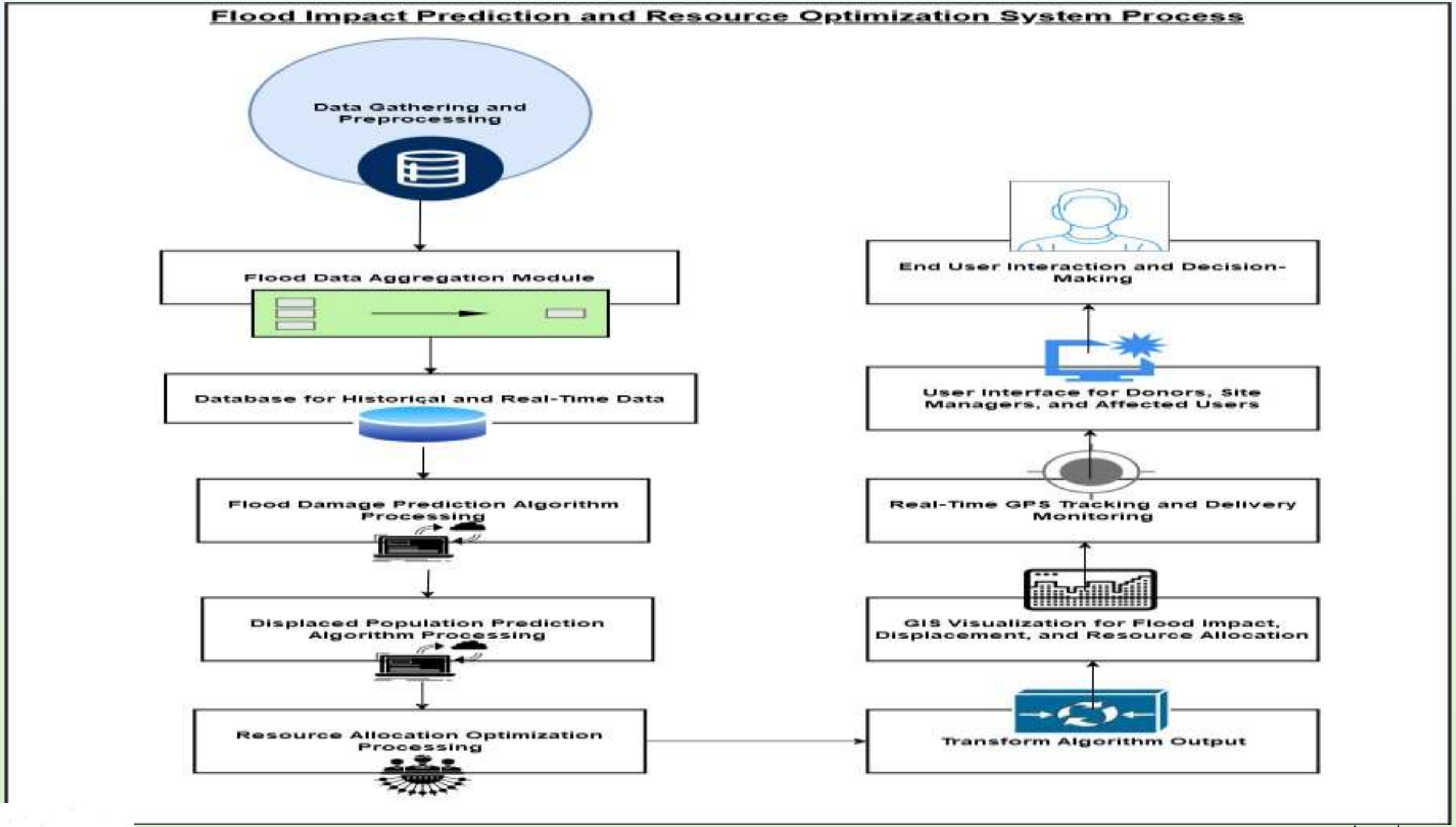
- How can a comprehensive, catchment-aware flood forecasting system be developed to
  - Predict flood related damages
  - Forecast displaced populations
  - Optimize resource allocation in Sri Lanka?



Specific Sub-Objectives

- **Flood Damage Prediction:** Develop a supervised learning model.
- **Displacement Forecasting:** Build a regression model to predict displaced populations.
- **Vehicle Tracking:** Implement GPS-enabled real-time tracking.
- **Resource Allocation Optimization:** Design an LP model for optimal resource distribution.
- **XAI Integration:** Enhance transparency and interpretability of predictions and allocations.

Methodology - System Diagram



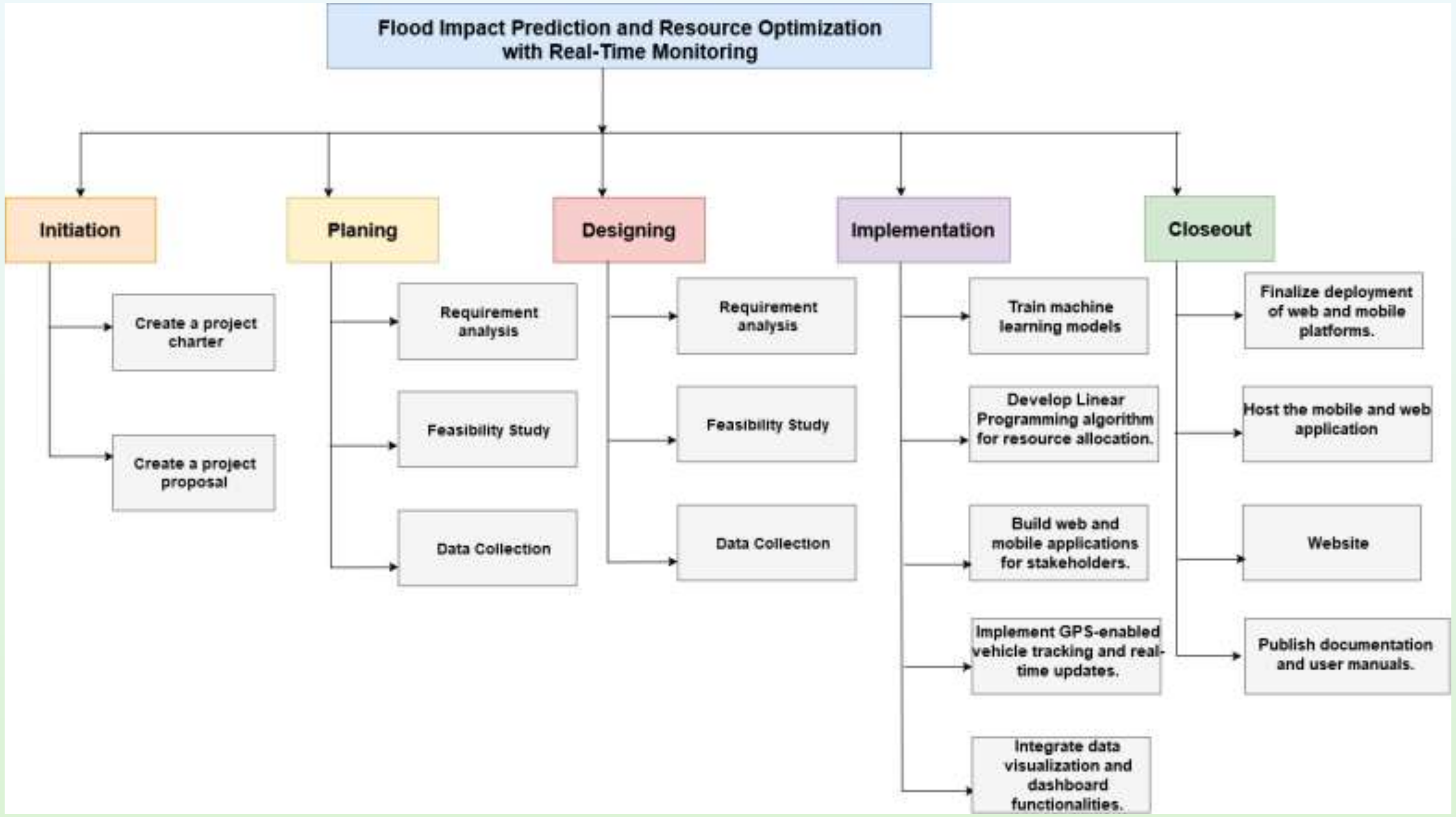
Technologies, Techniques, and Algorithms

- **Machine Learning Algorithms:**
  - **Random Forest Regressor:** For flood damage prediction & displacement forecasting.
  - **Linear Programming (LP):** For resource allocation optimization.
- **Explainable AI (XAI):**
  - For interpreting predictions and resource allocation decisions.
- **Technologies:**
  - **GIS Tools:** For analyzing flood-prone areas.
  - **Mobile & Web Apps:** For real-time tracking and resource management.
  - **GPS:** For live vehicle tracking.
- **Software Frameworks:**
  - **Python** (scikit-learn, pandas) for ML models.
  - **Flask/Django** for back-end development.
  - **React Native** for mobile app development.

Functional & Non-functional Requirements

- **Functional Requirements:**
  1. Predict flood damage levels and types.
  2. Forecast displaced populations.
  3. Optimize resource allocation.
  4. Provide real-time vehicle tracking.
  5. Integrate XAI for transparency.
- **Non-functional Requirements:**
  1. **Scalability**: Expandable to other flood-prone areas.
  2. **Reliability**: High uptime during disaster events.
  3. **Performance**: Real-time predictions and updates.
  4. **Usability**: User-friendly interface.
  5. **Security**: Safe data transmission and storage.

Work Breakdown Structure



References

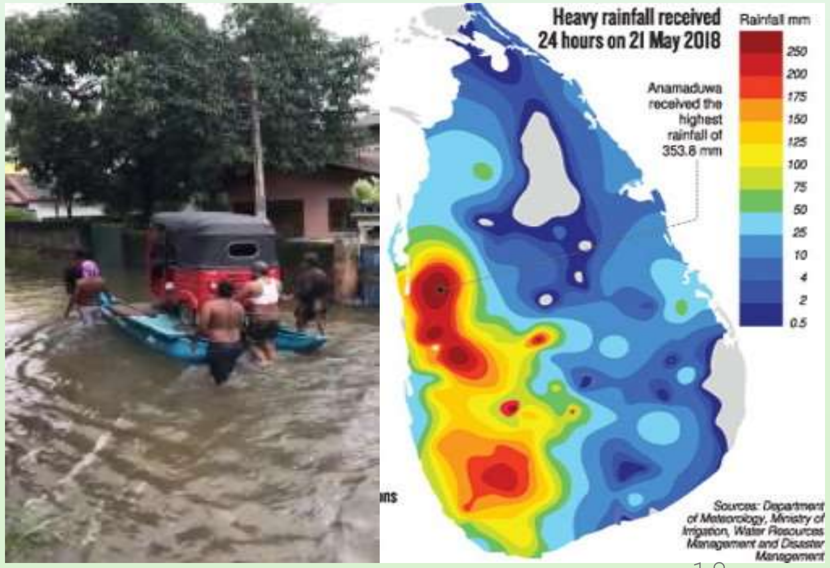
- [1] Mester, B., Willner, S. N., Frieler, K., & Schewe, J. (2021). Evaluation of river flood extent simulated with multiple global hydrological models and climate forcings. Environmental Research Letters, 16(9). Available: https://doi.org/10.1088/1748-9326/ac188d
- [2] A. N. Author, et al., "A Prediction Model for Minimization of Flood Effects using Machine Learning Algorithms," Journal Name, vol. X, no. Y, pp. Z-Z, 202X.
- [3] X. Y. Author, "Ensemble Learning Approaches for Predicting Flood Damage," Journal of Applied Computing, vol. 14, no. 3, pp. 123-134, 2023.
- [4] Mester, B., et al., "Societal Determinants of Flood-Induced Displacement," Nature Climate Change, 2024.

IT21340246 | Amarasekara S.K

Specialization: Information Technology

Introduction – Background

- Flooding is one of the most severe worldwide concerns, claiming millions of lives and incurring massive economic and infrastructure damage each year.
- Flood hazards are growing globally, exacerbated by climate change and increased urbanization, particularly in nations with distinct geographies such as Sri Lanka[1].
- System that addresses the complex issues of flood control by combining **Geographic Information Systems (GIS), IoT** and **ML** [2], [3].



Introduction - Background

- **Predicting flood-prone locations, efficiently visualizing data, determining safe routes for transportation, and producing trustworthy reports for sharing with authorities** are the objectives of the suggested system.
- The use of both historical and real-time data to identify flood-prone locations has shown great promise for machine learning techniques[4] ,[5].
- The Catchment-Aware Flood Forecasting System has the potential to transform flood risk management and improve resilience in areas that are vulnerable to flooding by using these cutting-edge technologies

Research Question

How can people find routes for transportation when passing through flood-prone areas ?

Any existing automated system to show the possibility of flooding , which area is at high risk in real time ?

Any existing automated system to show current flood-prone areas in real-time ?

Research Gap

Aspect	Existing Systems	Gap Identified
Socio-Economic Factors	There is no integrated system actionable and real-time forecasting.	This system provide dynamic forecasting and visualizations for better understandability.
Real-Time Data Integration and prediction	Use past data for predictions	Using past data and real-time data not only for prediction but also for visualization.
Alternative Routes Suggestion	No such system exist	Suggesting safe routes for vehicles to pass through flood-prone areas by navigating them with the help of map.
Flood Visualization	A manual procedure to visualize the flood	This study use flood prediction n equation

Specific and Sub-Objectives

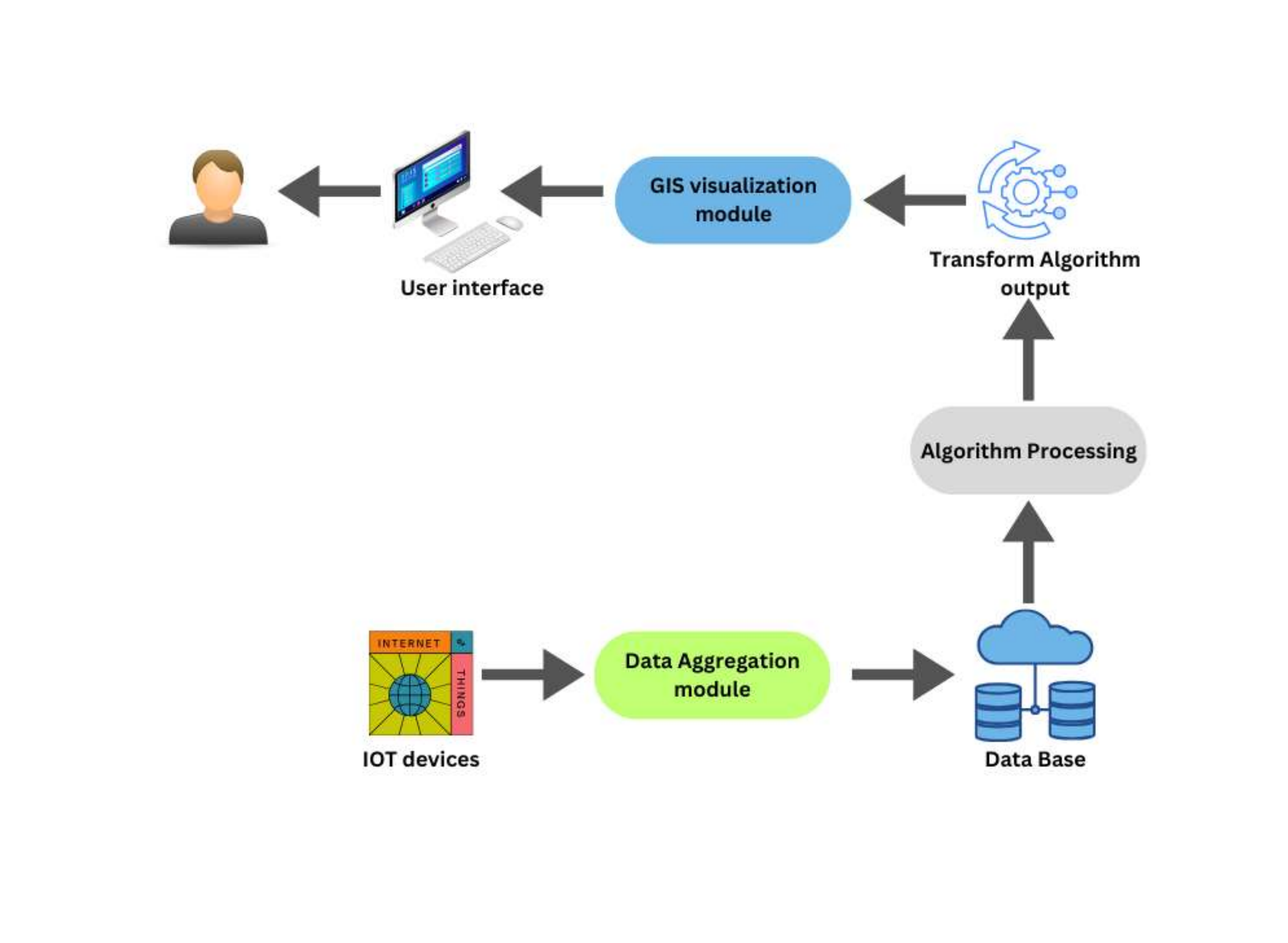
Develop an algorithm (Dijkstra's Algorithm) to identify alternative routes for the routes that are already affected by flood.

Suggesting safe routes for the people who wants to pass through flooded areas to minimize the risk and damage for transport facilities to ensure their day-to-day activities are not affected by flood.

Create predictions on flood affecting area from an equation with the use of some parameters as inputs which may cause flood occurrence use Choropleth Map visualization method .

Generating a trust worthy report for the people who are affected by flood to confirm their situation for the necessary authorities as needed.

Methodology – System Diagram



Functional, Non- Functional, and Software Requirements

Functional Requirements

- The System should accept data like rainfall, water level of the river and wind speed data and analyze using an algorithm to predict flooding areas.
- should be able to suggest alternative routes if existing routes got affected by flood.
- Integrate real-time data from IOT devices and synchronize them in real-time

Non-Functional Requirements

- System should handle increasing volume of data from IOT devices
- Provide a user-friendly interface.
- Secure authentication mechanism.
- The system should available at any moment for the users to access even at a extreme flood situation.

Software Requirements

- React Native
- Node Server
- Python
- Django REST
- Firebase
- JavaScript
- Mapbox Studio

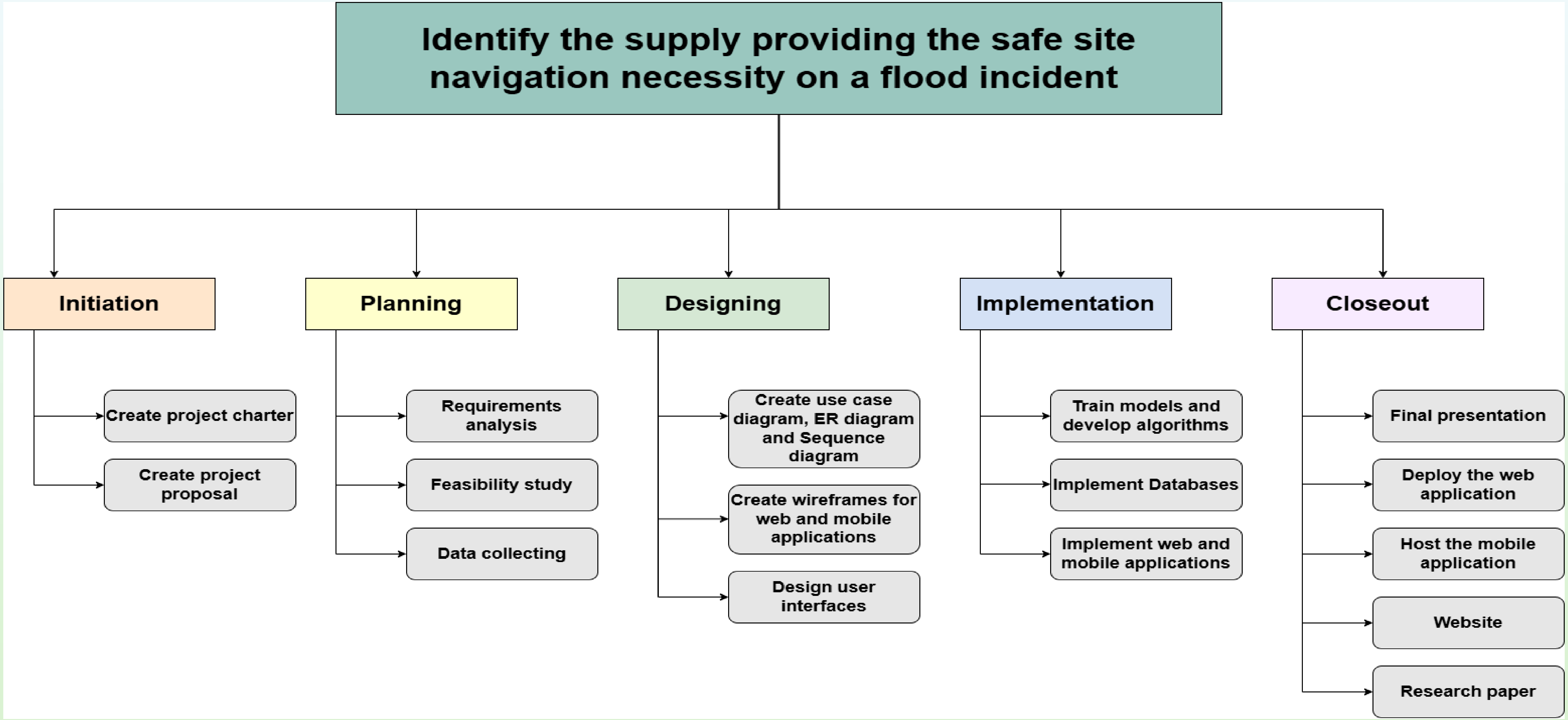
Methodology – Algorithms and Models

- **Machine Learning Algorithms**
  - Use equation to classify risk levels as high risk, medium risk, low risk and validate the equation from the relevant authorities
  - **Dijkstra's Algorithm** since it is focus on finding shortest path with smallest cumulative weight.
  - **Explainable AI** Provides transparency for navigation and decision-making, building user trust.

- **Technologies**
  - **GIS Tools** - Geospatial analysis to calculate the shortest path and safest route to each site(QGIS).
  - **GPS** - Use Google Maps API for turn-by-turn navigation to the nearest safe site
  - **QTTP/HTTP** protocols to send real-time data to server.
  - **Firebase** - Real-time Updates, **PostgreSQL** – Data storage

- **Software Frameworks**
  - **Django REST Framework**

Work Breakdown Structure



References

- [1] Sri Lanka Land Reclamation and Development Corporation, n.d. "Land Management and Flood-Prone Area Studies." [Online]. Available: http://www.sllrdc.lk/
- [2] A. Rahmati, S. Nazari Samani, R. Mahdavi, A. Pourghasemi, and A. Zeinivand, "Flood risk mapping using GIS and multi-criteria analysis: A case study," Environmental Earth Sciences, vol. 77, no. 23, p. 772, 2018.
- [3] B. Tomaszewski, Geographic Information Systems (GIS) for Disaster Management, 2nd ed. London, UK: Routledge, 2020.
- [4] T. A. M. Ali, P. Y. Yap, and K. Y. Lau, "Flood prediction using machine learning models: Literature review," Applied Sciences, vol. 10, no. 16, p. 5673, 2020.

Smith, C. Brown, and M. Wilson, "Flood disaster management:

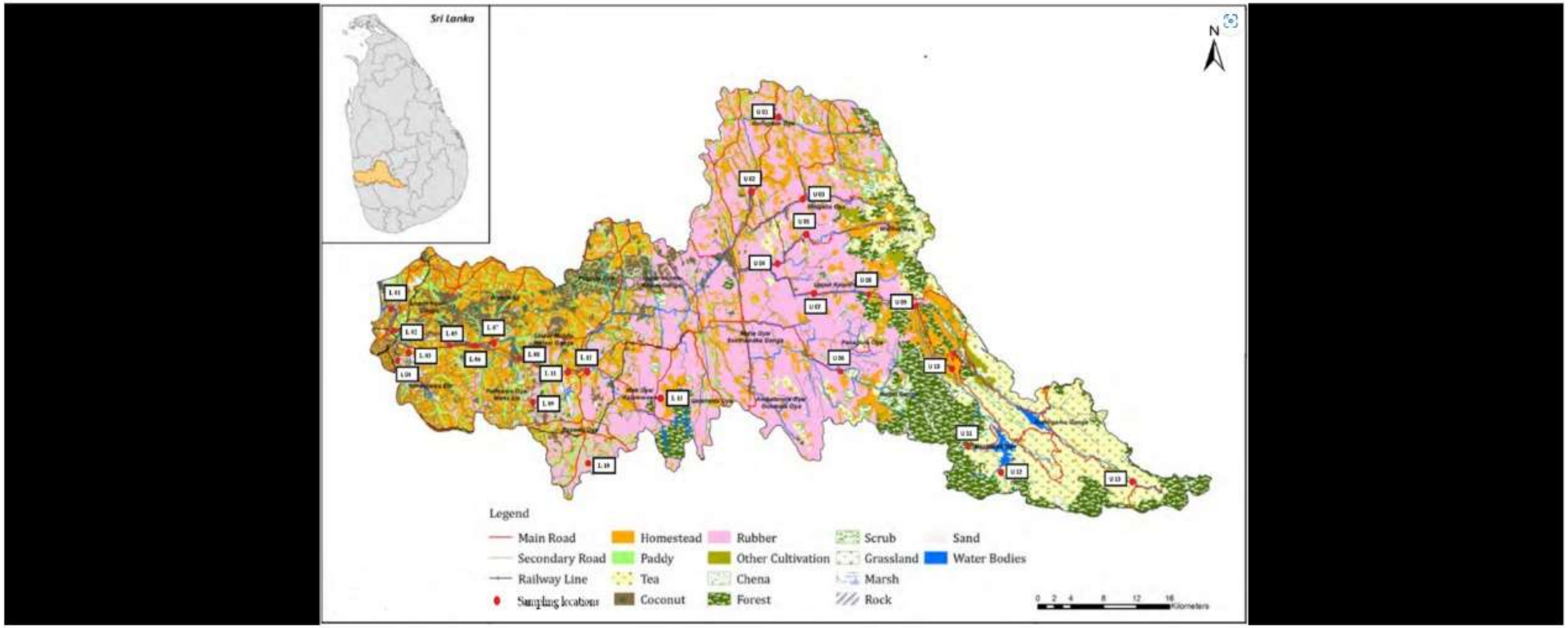
IT21816772 | Samaranayake N.A.S.R

Specialization: Information Technology

Introduction – Background

- The Internet of Things (IoT) enables seamless device connectivity and communication across industries
- IoT devices generate large volumes of data requiring efficient monitoring, management, and analysis.
- By addressing device health, maintenance, and system efficiency, the dashboard supports organizations in maintaining a competitive edge in the evolving IoT landscape.

Current data collecting Centers



Selected sampling locations of the upper and lower catchments of the Kelani River Basin (adapted from Edussuriya & Pathirage 2016). L01: Mattakkuliya, L02: Thotalanga, L03: Wellampitiya Bridge, L04: Kolonnawa, L05: Ambathale Bridge, L06: Biyagama, L07: Kaduwela, L08: Nawagamuwa, L09: Panagoda, L10: Padukka, L11: Hanwella Bridge, L12: Wak Oya, L13: Thummodara, U01: Lahupana Ella, U02: Kotiyakumbura, U03: Bulathkohupitiya, U04: Parussalla, U05: Wee Oya, U06: Panakoora, U07: Alagal Oya, U08: Kithulgala, U09: Kalugala Bric

Research Gap

Aspect	Existing Systems	Gap Identified
Predictive Maintenance & AI Integration	Many dashboards rely on reactive maintenance .	Limited research on incorporating advanced AI/ML algorithms to anticipate failures before they happen.
Human-Computer Interaction	Some dashboards can be complex and difficult for non-technical users to understand and operate.	Limited research on optimizing the user interface and experience for different roles

Specific and Sub-Objectives

- To design and implement an IoT dashboard for monitoring connected devices in real-time.
- To assess user requirements for administrative tasks in IoT system management.
- To analyze common device malfunctions and develop predictive maintenance strategies
- To measure the operational efficiency improvements achieved through IoT systems.

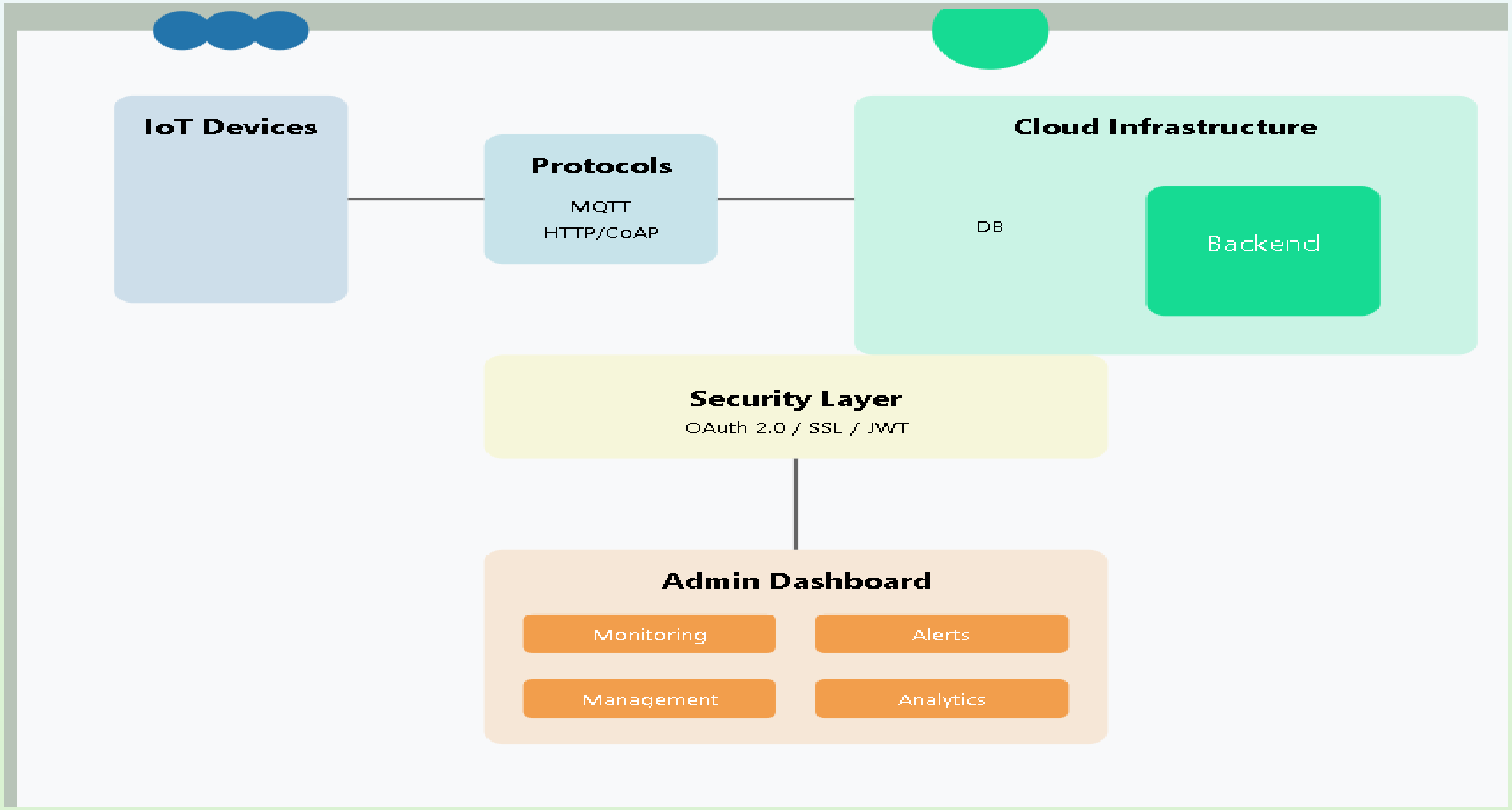
Research Question

How can an admin dashboard optimize real-time monitoring of IoT devices in a heterogeneous ecosystem?

What features are essential for a user-friendly IoT maintenance admin dashboard?

How can machine learning models be integrated into an IoT dashboard to predict device failures?

Methodology – System Diagram



Used techniques and Technologies

Technique s

- Sensor Techniques
- Communication Techniques
- Cloud Integration
- Alert Systems

Technologi es

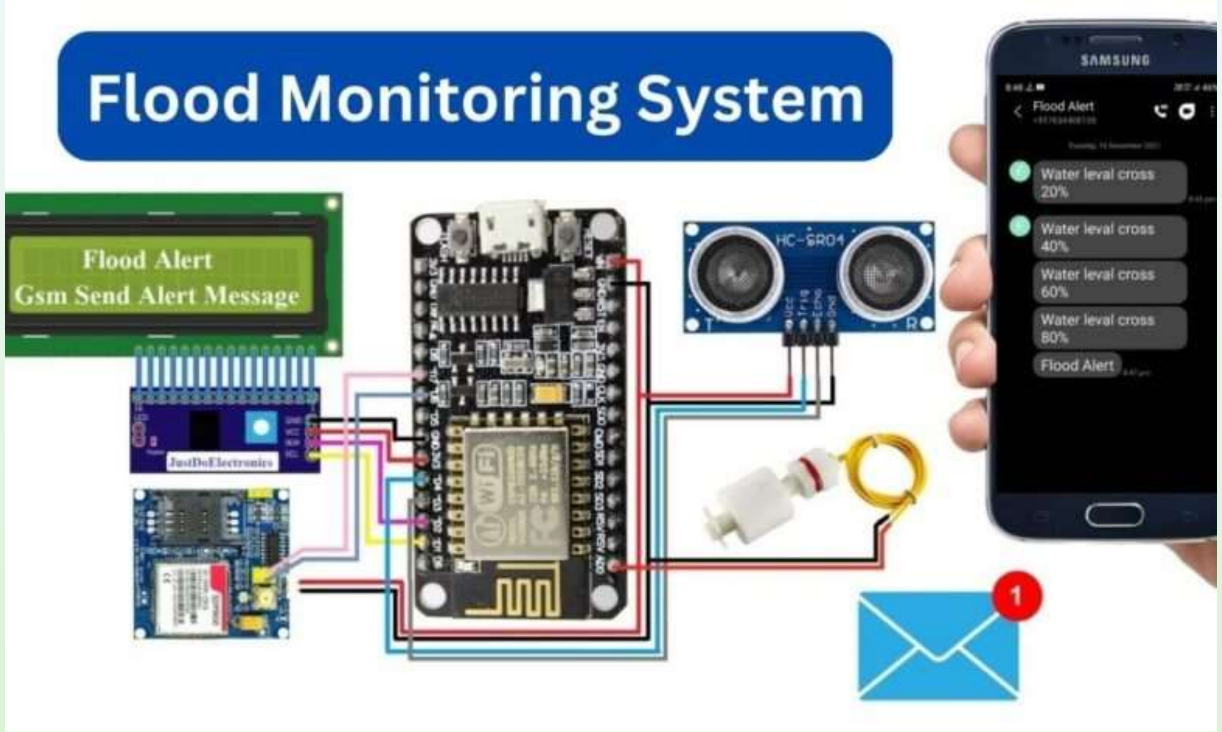
- Arduino UNO / Nano
- Firebase
- Mobile Notifications

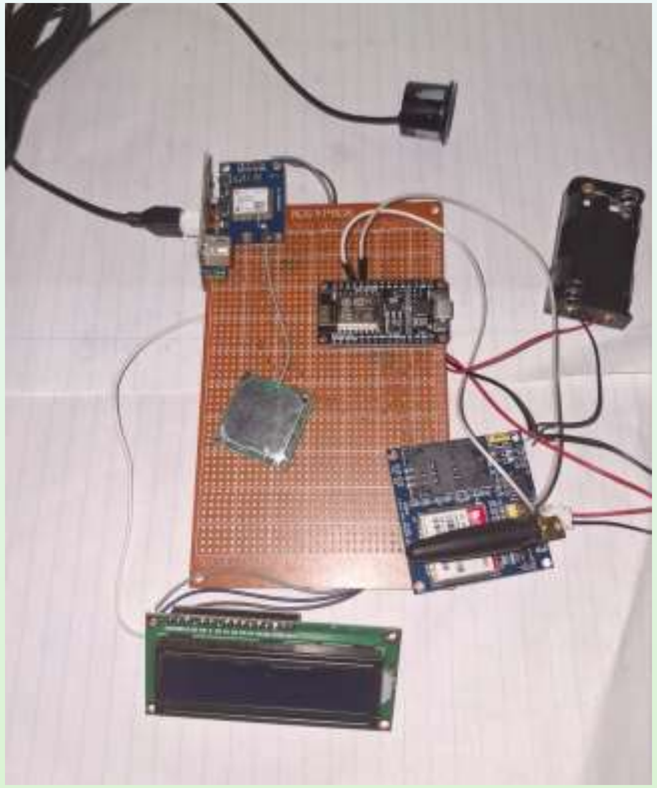
Methodology

Evidence of Completion

- Component Selection
- System Design and Development
- Implementation
- Testing and Evaluation
- Circuit setup
- LCD showing real-time water levels

Component Selection
Evidence of Completion





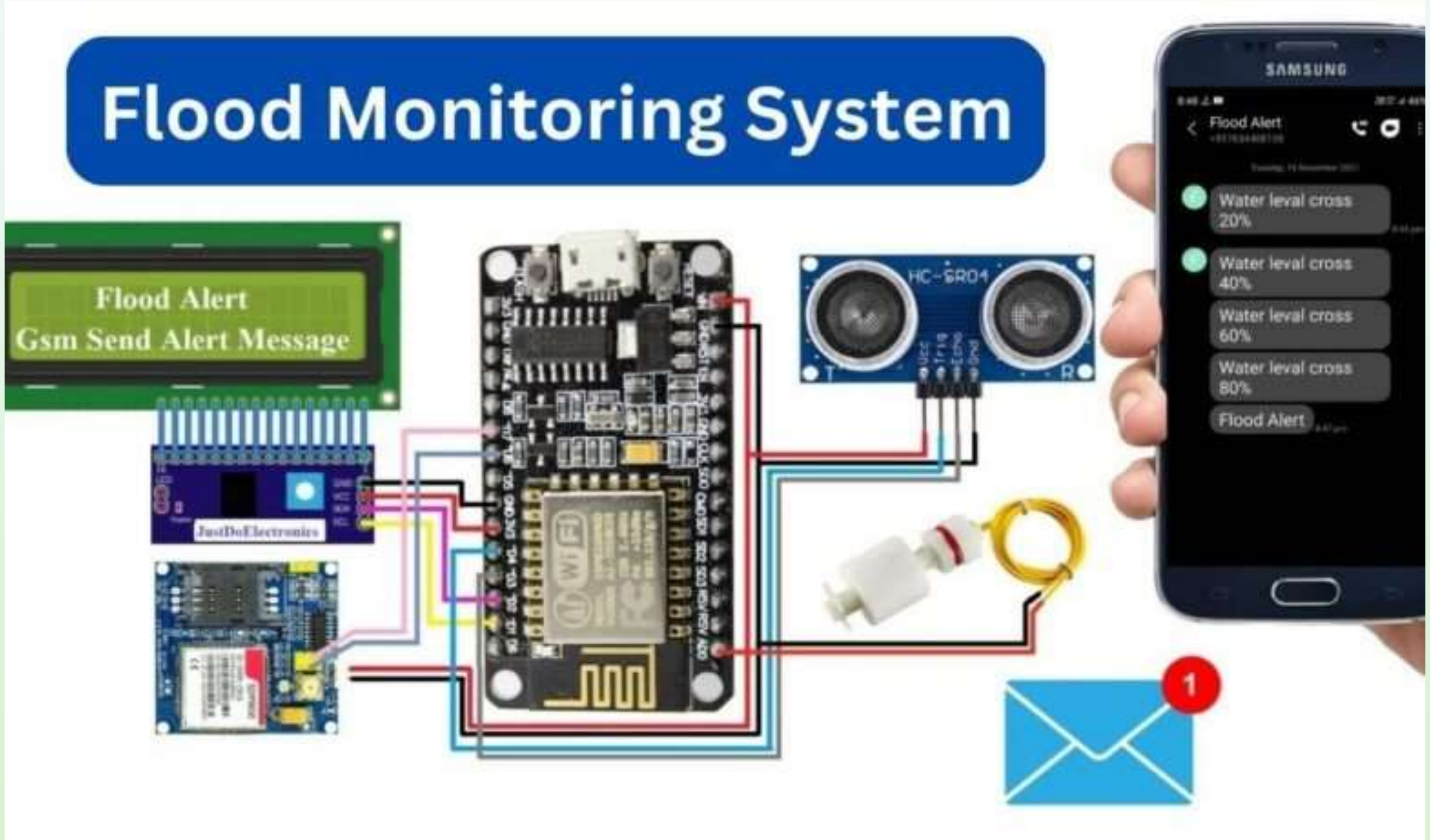
System Design and Development

Evidence of Completion

```
void loop() {

  ultersonic();    // Read sensor values

  if (buttonState == HIGH) {
    // Water level is HIGH
    Serial.println("WATER LEVEL -HIGH");
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("  HIGH ALERTS");
    lcd.setCursor(0, 1);
    lcd.print("WATER LEVEL-");
    lcd.setCursor(13, 1);
    lcd.print(distance1);

  } else {
    // Water level is LOW
    Serial.println("WATER LEVEL - LOW");
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("   LOW ALERTS");
    lcd.setCursor(0, 1);
    lcd.print("WATER LEVEL-");
    lcd.setCursor(13, 1);
    lcd.print(distance1);
  }
```

Circuit setup

Evidence of Completion



Functional, Non- Functional, and Software Requirements

Functional Requirements

- Device Management
- Real-Time Monitoring
- Predictive Maintenance
- Alerts and Notifications
- Maintenance Logs and History
- User Authentication & Roles
- Dashboard Customization

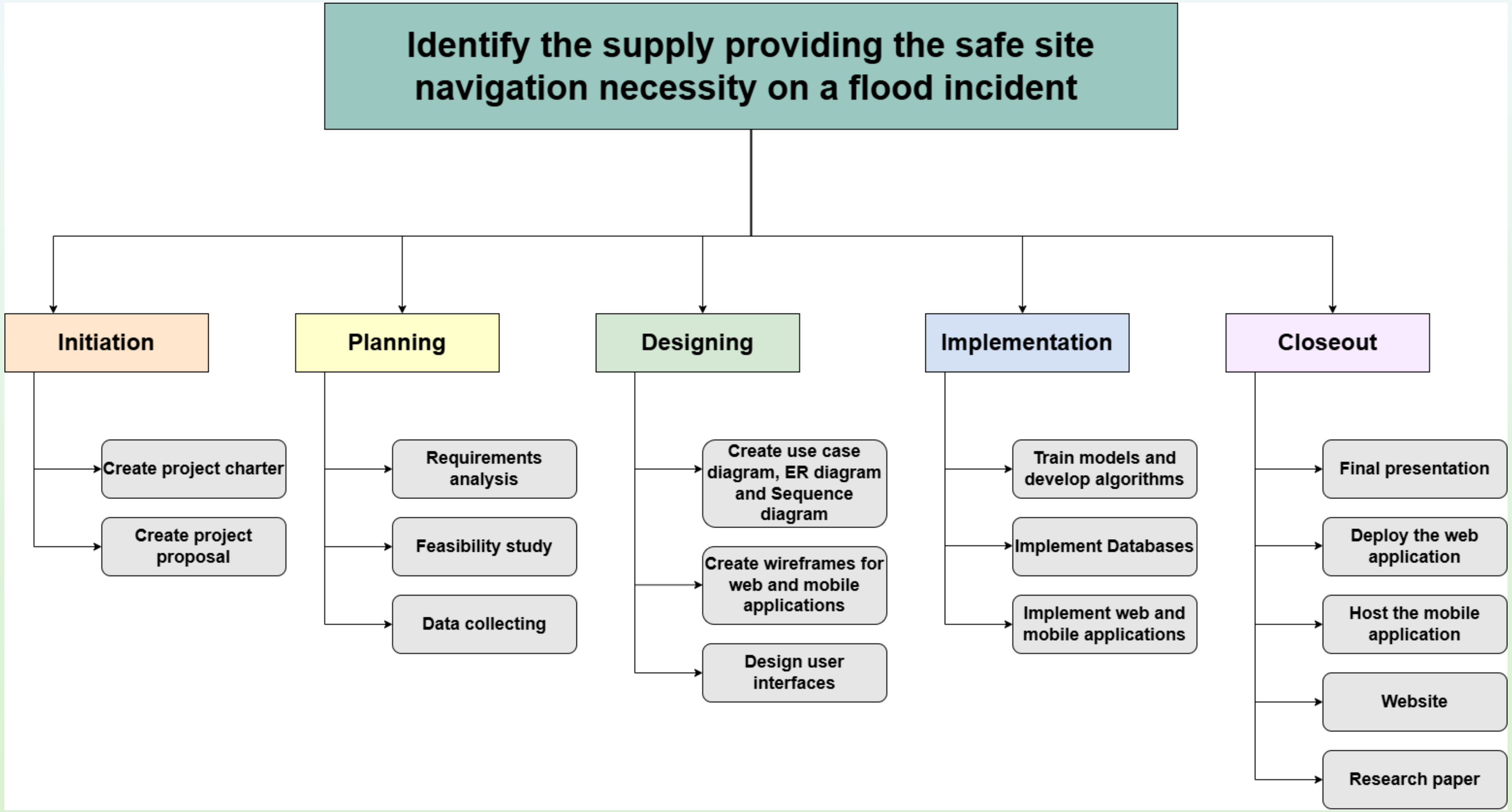
Non-Functional Requirements

- Performance
- Reliability and Availability
- Scalability
- Security
- Usability
- Data Integrity
- Compliance
- Maintainability

Software Requirements

- Angular/React.js
- Node.Js
- MongoDB
- MQTT
- PyTorch
- AWS IoT Core

Work Breakdown Structure



Completion and Future Work

Completion of the component

50% of the project is completed with creating device

Future implementation

In the 50%-90% of the progress, hoping to improve the Device with float sensor an

Complete the Mobile application development

Complete the other sub-objectives

References

- **Hydrology and Disaster Management Division, Irrigation Department**
- Evaluation of water quality in the upper and lower catchments of the Kelani River Basin, Sri Lanka

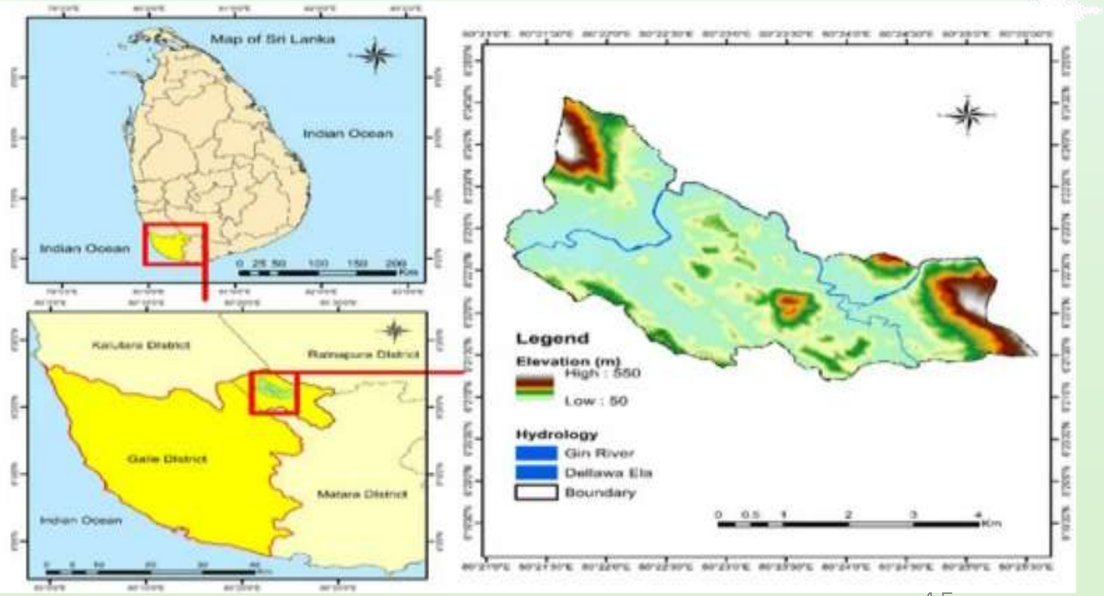
  Chandima Narangoda;Deeptha Amarathunga; Chandima Deepani Dangalle
- Navy(https://floodms.navy.lk/wlrs/index.php)

IT21305214 | Dias A.H.S.G

Specialization: Information Technology

Introduction - Background

- Advanced flood forecasting and management systems are required due to the growing frequency and intensity of floods brought on by urbanization and climate change [1].
- This system integrates contemporary technologies like **artificial intelligence, geographic mapping, and predictive analytics** to handle the complex issues of flood preparedness and response [2].



Introduction - Background

- Many existing solutions focus on flood prediction and monitoring but lack comprehensive features for ensuring user safety, such as **personalized navigation to safe sites, real-time updates on supply availability, and two-way communication for support.**
- For mapping purposes, this component uses user position monitoring, predictive analytics to alert users to possible dangers, and algorithms to evaluate flood risks [3].
- Recent studies emphasize the importance of crowdsourced data in monitoring resource availability at relief centers. For instance, [4] proposed a framework that leverages IoT sensors and mobile applications to provide real-time updates on **essential supplies, including food, water, and medical resources.**

Research Question

Is there a possible way to predict the diseases can spread and recommend the required essentials?

How can affected people find their essential supplies??

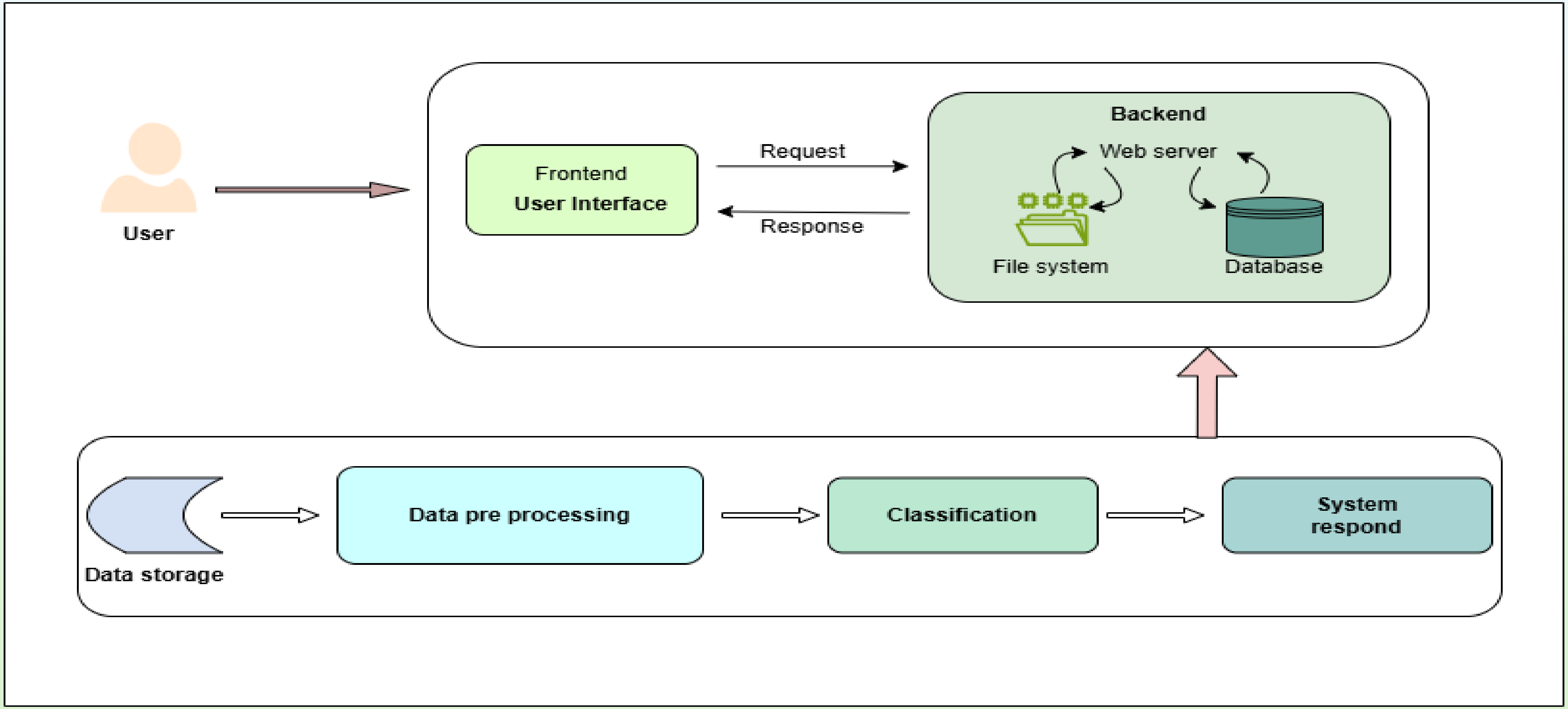
Research Gap

Aspect	Existing Systems	Gap Identified
Socio-Economic Factors	There is no integrated system actionable and real-time updates about safe site statuses (Required essential supplies)	Sending Alerts About Nearby Flood Risks or Safe site updates
Predict the diseases due to flood and recommend the required essentials	No such system exists	Train a model to recommend required essentials based on the flood risk
Supply requesting and providing options	No such system exists	Connect with relief centers to collect supply data (e.g., food, water, medicine) via APIs or manual entry.
Suggesting the Safest [illegible]earby Sites to	A manual procedure provides supplies	Use multi-criteria decision-making

Specific and Sub-Objectives

- To provide security guidance to flood-affected individuals, ensuring their safety and well-being during flood events .
- To establish two-way communication mechanisms for users to provide feedback, request supplies, and receive support during flood events.
- To create novelty features, including dynamic suggestions for safe locations based on the risk, and to predict the required essentials for diseases that can spread due to flood risk
- To enable updates on safe site statuses, supply availability, calculate required supplies for safety centers.

Methodology – System Diagram



Used techniques and Technologies

Techniques

- Supervised Learning
- Data encoding
- Feature engineering
- Model training
- Evaluation

Technologies

- Python
- Pandas
- Scikit-learn
- Tensorflow
- Google Colab

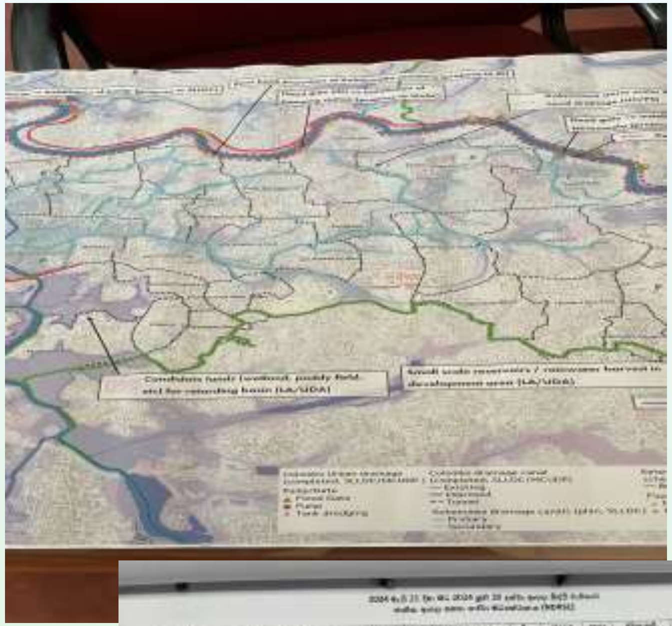
Methodology

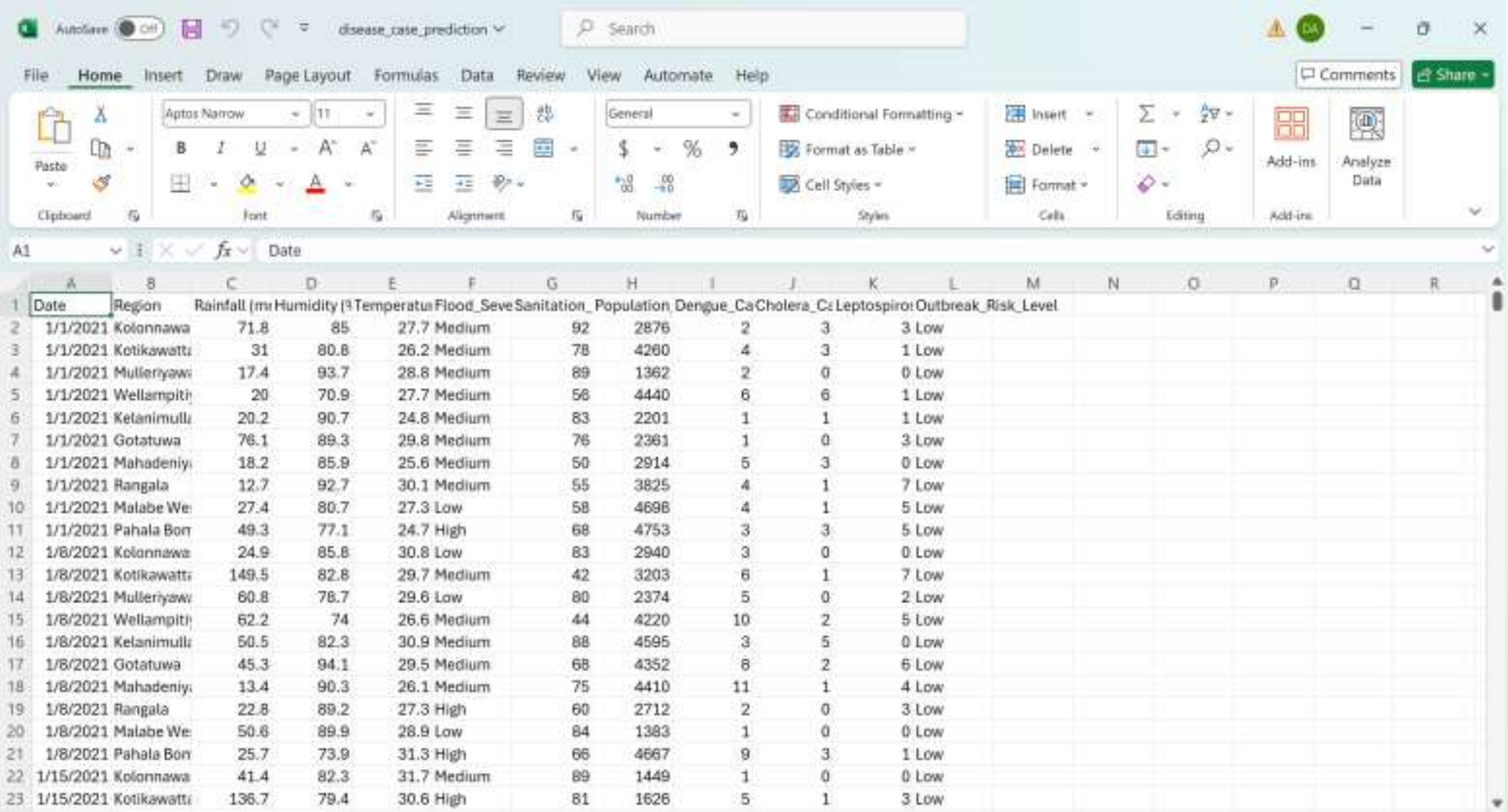
Evidence of Completion

- **Data collection**
- **Data analysis**
- **Data Preprocessing**
- **Serialization**
- **Host the finalized model**
- **Display the results in a simple UI**

Data collection and Preprocessing

Evidence of Completion



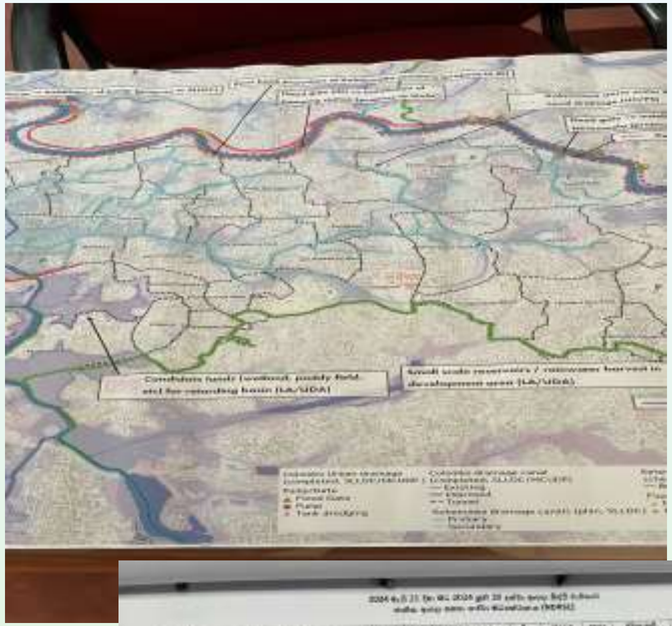


Date	Region	Rainfall (mı	Humidity (ᵑ	Temperatu	Flood_Seve	Sanitation_	Population	Dengue_Ca	Cholera_Ca	Leptospiro	Outbreak_Risk_Level
1/1/2021	Kolonnawa	71.8	85	27.7	Medium	92	2876	2	3	3	Low
1/1/2021	Kotikawatt	31	80.8	26.2	Medium	78	4260	4	3	1	Low
1/1/2021	Mulleriyaw	17.4	93.7	28.8	Medium	89	1362	2	0	0	Low
1/1/2021	Wellampiti	20	70.9	27.7	Medium	56	4440	6	6	1	Low
1/1/2021	Kelanimull	20.2	90.7	24.8	Medium	83	2201	1	1	1	Low
1/1/2021	Gotatuwa	76.1	89.3	29.8	Medium	76	2361	1	0	3	Low
1/1/2021	Mahadeniy	18.2	85.9	25.6	Medium	50	2914	5	3	0	Low
1/1/2021	Rangala	12.7	92.7	30.1	Medium	55	3825	4	1	7	Low
1/1/2021	Malabe We	27.4	80.7	27.3	Low	58	4696	4	1	5	Low
1/1/2021	Pahala Bor	49.3	77.1	24.7	High	68	4753	3	3	5	Low
1/8/2021	Kolonnawa	24.9	85.8	30.8	Low	83	2940	3	0	0	Low
1/8/2021	Kotikawatt	149.5	82.8	29.7	Medium	42	3203	6	1	7	Low
1/8/2021	Mulleriyaw	60.8	78.7	29.6	Low	80	2374	5	0	2	Low
1/8/2021	Wellampiti	62.2	74	26.6	Medium	44	4220	10	2	5	Low
1/8/2021	Kelanimull	50.5	82.3	30.9	Medium	88	4595	3	5	0	Low
1/8/2021	Gotatuwa	45.3	94.1	29.5	Medium	68	4352	8	2	6	Low
1/8/2021	Mahadeniy	13.4	90.3	26.1	Medium	75	4410	11	1	4	Low
1/8/2021	Rangala	22.8	89.2	27.3	High	60	2712	2	0	3	Low
1/8/2021	Malabe We	50.6	89.9	28.9	Low	84	1383	1	0	0	Low
1/8/2021	Pahala Bor	25.7	73.9	31.3	High	66	4667	9	3	1	Low
1/15/2021	Kolonnawa	41.4	82.3	31.7	Medium	89	1449	1	0	0	Low
1/15/2021	Kotikawatt	136.7	79.4	30.6	High	81	1626	5	1	3	Low

Finalized – Created dataset

Data collection and Preprocessing

Evidence of Completion



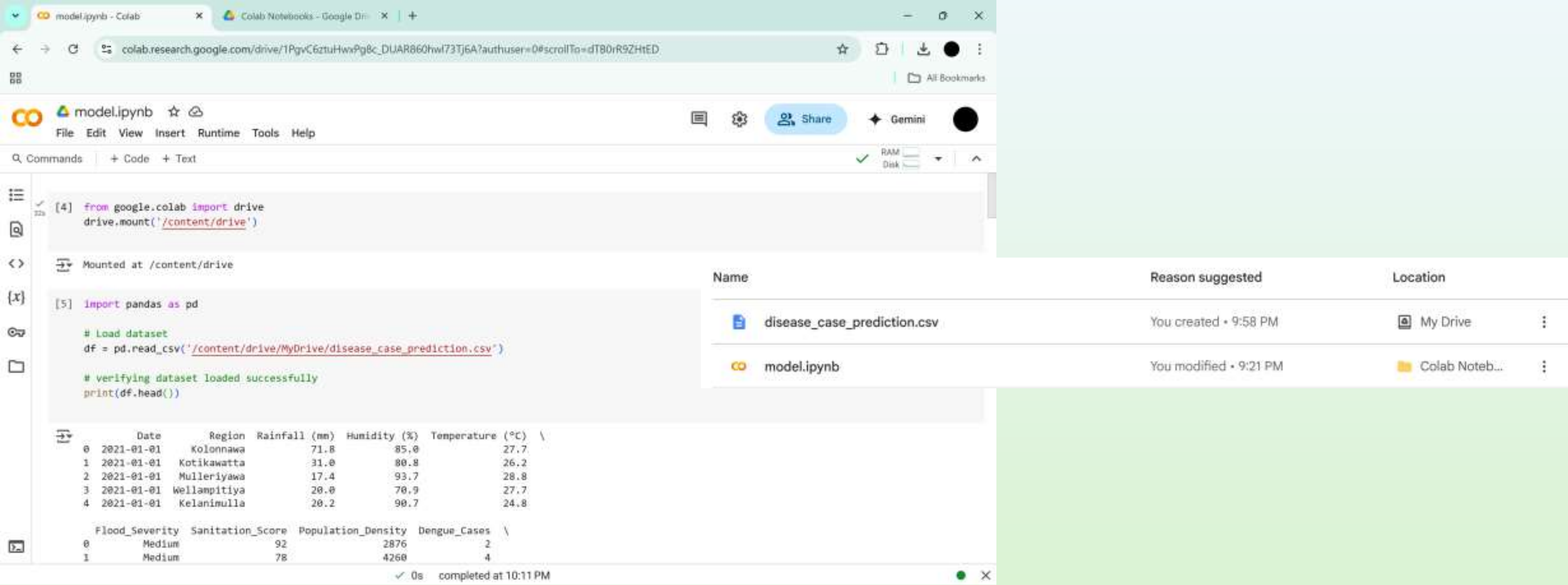
```python
import pandas as pd
```

```python
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import classification_report

# Encode categorical variables
le_region = LabelEncoder()
le_severity = LabelEncoder()
le_risk = LabelEncoder()

df["Region"] = le_region.fit_transform(df["Region"])
df["Flood_Severity"] = le_severity.fit_transform(df["Flood_Severity"])
df["Outbreak_Risk_Level"] = le_risk.fit_transform(df["Outbreak_Risk_Level"])
```

Data preprocessing implementation

Model training

Evidence of Completion



Model training on Google Colab and saved to Drive

Supervised Learning Architecture

Evidence of Completion

```python
# Define features and targets
features = ["Region", "Rainfall (mm)", "Humidity (%)", "Temperature (°C)",
            "Flood_Severity", "Sanitation_Score", "Population_Density"]

X = df[features]
```

Input features from the dataset

```python
from sklearn.ensemble import RandomForestRegressor, RandomForestClassifier
```

```python
# Splitting data
X_train, X_test, yd_train, yd_test = train_test_split(X, y_dengue, test_size=0.2, random_state=42)
_, _, yc_train, yc_test = train_test_split(X, y_cholera, test_size=0.2, random_state=42)
_, _, yl_train, yl_test = train_test_split(X, y_lepto, test_size=0.2, random_state=42)
X_train_risk, X_test_risk, yr_train, yr_test = train_test_split(X_risk, y_risk, test_size=0.2, random_state=42)

# Train regression models
model_dengue = RandomForestRegressor(random_state=42).fit(X_train, yd_train)
model_cholera = RandomForestRegressor(random_state=42).fit(X_train, yc_train)
model_lepto = RandomForestRegressor(random_state=42).fit(X_train, yl_train)

# Train classifier
classifier = RandomForestClassifier(random_state=42).fit(X_train_risk, yr_train)
```

Training the model

Why is Supervised Learning Architectu

- **Have labeled training data** (inputs + correct outputs)

  Inputs: Rainfall, Humidity, Region, Flood Severity, etc.
  Outputs: Dengue cases, Cholera cases, Outbreak Risk Level, etc.

- Prediction Is the Goal

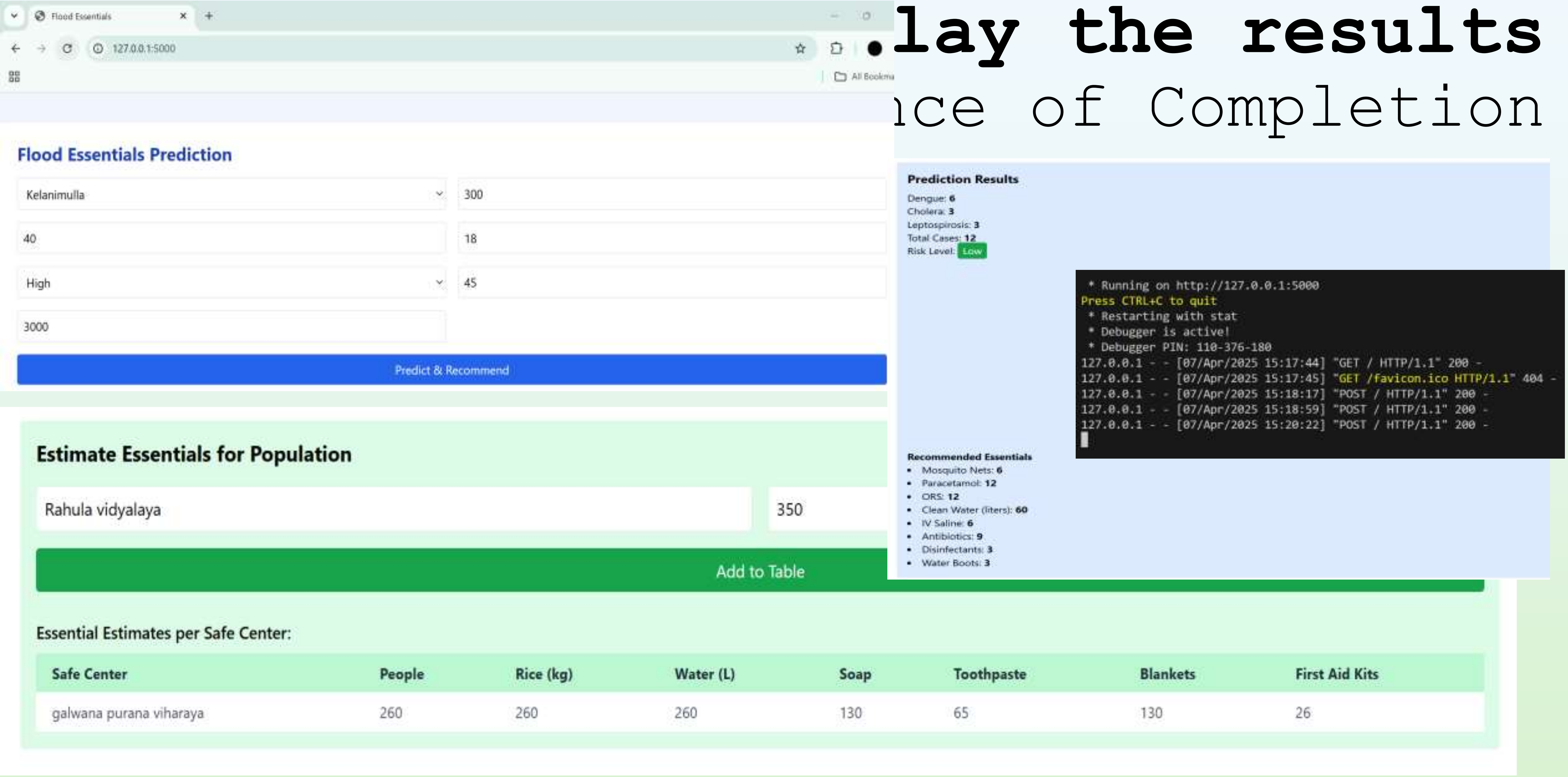
  Number of disease cases (regression)
  Risk level category (classification)

- **Tabular Data Structure**

  Supervised learning algorithms like those in **scikit-learn** are perfect for **structured/tabular data** (e.g., CSV format),

lay the results

nce of Completion



Functional, Non- Functional, and Software Requirements

Functional Requirements

- Monitor the requirements of essential supplies at the sites.
- The system should predict the diseases and the recommended essentials based on weather forecasts and floods.
- The system should enable user location tracking for personalized guidance.
- Suggest the safest and nearest sites dynamically.
- Identify and update the status of nearby safe locations dynamically.

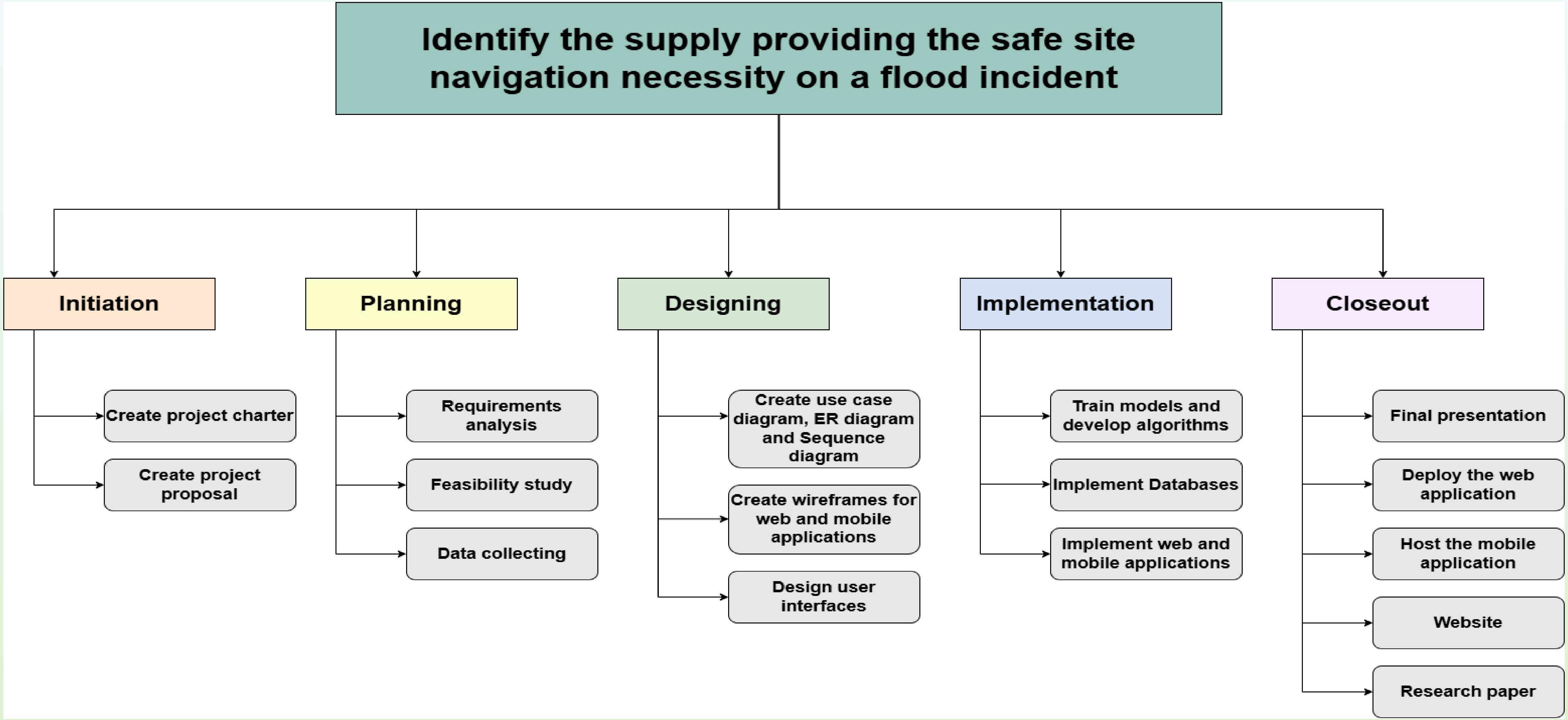
Non-Functional Requirements

- Guarantee 24/7 uptime during critical periods like floods.
- Provide a user-friendly interface.
- Protect user data and communication from unauthorized access.
- Ensure the system can handle a large number of users during disaster scenarios.

Software Requirements

- React Native
- Node Server
- Python
- Firebase
- JavaScript
- PostgreSQL
- Google Colab

Work Breakdown Structure



Completion and Future Work

Completion of the component

- ✓ 50% of the project is completed with the trained model with a novel objective
- ✓ Displaying the recommended essential supplies based on predicted disease results due to the flood
- ✓ Estimated essentials for the affected population at the safety centers

Future implementation

- ✓ In the 50%-90% of the progress, hoping to improve the model further with a stronger dataset, and also to train the model using the Random Forest model
- ✓ Complete the Mobile application development
- ✓ Complete the other sub-objectives, including Supply availability up-to-date and alerts, User communication and feedback system, and Users can request their required supplies and support

References

- [1] W. M. O. (WMO), " Guidelines on Flood Forecasting and Warning Systems.," World Meteorological Organization (WMO), 2021.
- [2] J. &. B. L. Smith, "Advances in Flood Forecasting and Disaster Management. Springer," Advances in Flood Forecasting and Disaster Management. Springer, 2020.
- [3] D. e. a. Chen, "Real-time Flood Forecasting Using Machine Learning Techniques," Journal of Hydrology, 588, 125036, 2020.
- [4] A. e. a. Kumar, "IoT-enabled Resource Management for Flood Relief Operations," 2021.

THANK YOU